

UNIVERSIDAD FRANCISCO DE VITORIA

ESCUELA POLITÉCNICA SUPERIOR

COMPUTER SCIENCE DEGREE

FINAL DEGREE PROJECT

RESEARCH TRACK

Sign Lenguage Recognition System: Hand Detection and Word Classification using Deep Learning

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CALIFICACIÓN DEL PROYECTO DE FIN DE GRADO

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Lugar y fecha: Pozuelo de Alarcón, a ____ de _____ de 2026

Agradecimientos

Aquí se incluirían los agradecimientos personales y profesionales. No olvidarse de agradecer la ayuda recibida, especialmente si se ha realizado el proyecto gracias a una beca, contrato o algún tipo de subvención o apoyo.

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Resumen

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Palabras claves: reconocimiento de lengua de signos, detección de manos, aprendizaje profundo, YOLO, detección de objetos, visión por computador, ASL

Abstract

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Keywords: sign language recognition, hand detection, deep learning, YOLO, object detection, computer vision, ASL

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Chapter 1

Introduction

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1.1 Problem description

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1.2 Outline

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Chapter 2

State of the Art

2.1 From Symbolic to Subsymbolic AI

Early Artificial Intelligence (AI) systems were symbolic: they represented knowledge using explicit rules and logic, and solved problems through handcrafted if/then decision trees. This paradigm achieved success in constrained domains, but it struggled with perception tasks such as vision and speech because the real world is noisy, ambiguous, and high-dimensional. The so-called "knowledge acquisition bottleneck" meant that as tasks grew in complexity, engineering and maintaining rule bases became impossible. In the context of sign language, symbolic systems could encode simple gesture prototypes but failed to capture the fine-grained variability across signers, viewpoints, and execution speeds.

The shift to subsymbolic AI reframed intelligence as pattern recognition. Instead of encoding expert rules, models learn distributed representations from data. This change was prompted by advances in statistical learning and neural networks, culminating in deep learning methods that learn hierarchical features directly from raw inputs [1]. Subsymbolic models are particularly effective in perception because they optimize representations for the task objective, allowing subtle visual patterns to emerge without explicit programming. In sign language recognition, subsymbolic models learn the visual semantics of hand shapes, motions, and temporal dynamics from examples rather than hand-coded descriptors.

From a theoretical standpoint, subsymbolic approaches replace symbolic inference with continuous function approximation. A deep network implements a parameterized mapping f_θ that transforms inputs into task-specific embeddings, optimized by minimizing a loss over data. This paradigm aligns with the "computational" and "algorithmic" levels of explanation in vision, where representation learning is the core challenge. By training end-to-end models on large datasets, modern systems discover invariances to lighting, background, and signer variability that are nearly impossible to enumerate symbolically.

The subsymbolic shift also enabled modular pipelines in which perceptual frontends (e.g., hand pose estimation) feed compact, structured features to temporal models. This decoupling is powerful for SLR: the vision module learns spatial representations, while the sequence model learns dynamics. The result is a scalable, data-driven framework that has largely replaced rule-based approaches in

the literature. Contemporary SLR therefore reflects the broader AI trajectory: from symbolic rules to learned representations and, increasingly, to architectures that encode inductive biases (e.g., convolutional and recurrent structures) while remaining fully trainable from data.

2.2 Feature Extraction and Pose Estimation

The first challenge in Sign Language Recognition is extracting meaningful spatial features from video frames. This field has evolved from processing raw pixels to extracting high-level geometric structures.

2.2.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) represent the foundational technology for modern computer vision [1]. Unlike fully connected networks that treat inputs as flat vectors, CNNs preserve the spatial structure of images through three main operations: convolution, non-linearity, and pooling. A typical convolutional layer computes a feature map $\mathbf{y}$ as:

$$\mathbf{y}_{i,j,k} = \sigma\left(\sum_{c=1}^{C}\sum_{u,v}\mathbf{w}_{u,v,c,k}\,\mathbf{x}_{i+u,j+v,c} + b_k\right) \tag{2.1}$$

where $\mathbf{x}$ is the input tensor, $\mathbf{w}$ are learned kernels (filters), and σ is a non-linearity (e.g., ReLU). This operation introduces an inductive bias toward *translation equivariance*, meaning the network can recognize features regardless of their position in the image.

Deep CNN architectures, such as AlexNet [2] or ResNet, stack many such layers to learn hierarchical representations. Initial layers typically behave as Gabor filters, detecting simple edges and textures. Deeper layers combine these low-level features to detect complex shapes, and finally, semantic objects like hands or faces. The use of pooling layers further enhances robustness by introducing local translation invariance, allowing the network to detect features even if they are slightly shifted or distorted.

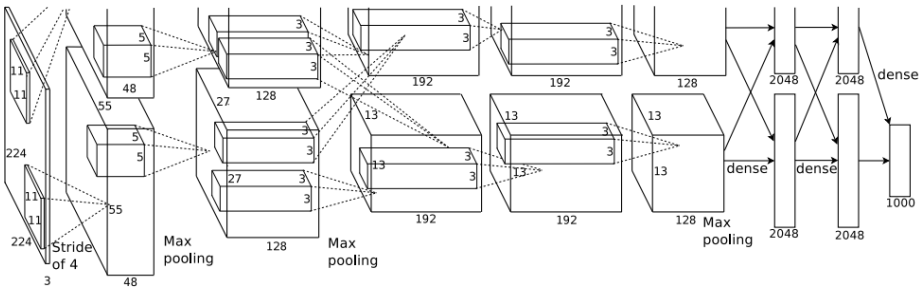


Figure 2.1: Architecture of AlexNet. Note the split processing paths originally designed for dual-GPU training. Source: Krizhevsky et al. [2].

2.2.2 From Pixels to Landmarks: Pose Estimation

While CNNs can process raw RGB pixels directly (pixel-based approach), this method is computationally expensive and sensitive to background noise and lighting conditions. To address this, *Pose Estimation* techniques leverage CNNs to regress the coordinates of anatomical keypoints, transforming the visual problem into a geometric one [3]. In practice, pose pipelines are commonly framed as either top-down (detect hands first, then regress keypoints) or bottom-up (predict keypoints first, then group them into hands). For SLR, the top-down approach is especially attractive because it focuses computation on the hand region and naturally supports frame-to-frame tracking.

MediaPipe Hands [4] is a seminal system in this domain. It employs a two-stage pipeline to achieve real-time performance:

1. **Palm Detector:** A lightweight single-shot detector operates on the full image to locate the hand palm. This step is crucial because detecting a rigid object like a palm is simpler than detecting articulating fingers.
2. **Landmark Regressor:** A high-capacity CNN processes the cropped hand region to predict the 3D coordinates of 21 skeletal joints.

This architecture decouples detection from regression, allowing the system to track hands efficiently by re-using the crop from the previous frame. For SLR, this provides a compact, invariant representation of the hand shape, filtering out irrelevant visual information such as background clutter or skin tone variations. The landmark representation also enables geometric normalization (e.g., translation and scale alignment) and the derivation of higher-level features such as bone vectors, joint angles, and inter-finger distances, which often yield more stable temporal dynamics than raw pixels.

Pose-based pipelines frequently integrate additional cues beyond the hands. Upper-body or face landmarks can capture non-manual signals (head pose, facial expression) that disambiguate signs or encode grammatical structure. Even when only hand keypoints are used, modern systems typically propagate per-joint confidence scores to handle occlusions and motion blur, masking unreliable points rather than forcing the sequence model to fit noise. These design choices make the pose representation more robust to real-world variability in signer appearance, speed, and viewpoint.

2.3 Sequence Modeling

Since sign language is inherently temporal, recognizing signs requires modeling the evolution of features over time. Standard feedforward networks cannot handle this task because they accept fixed-size inputs and have no memory of previous states. This limitation necessitates the use of Recurrent Neural Networks (RNNs).

2.3.1 Recurrent Neural Networks (RNNs)

RNNs introduced the concept of "memory" in neural networks by maintaining a hidden state $\mathbf{h}_t$ that is updated at each time step based on the current input $\mathbf{x}_t$ and the previous state $\mathbf{h}_{t-1}$:

$$\mathbf{h}_t = \phi(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h) \tag{2.2}$$

This recurrent connection allows the network to process sequences of variable length, sharing the same parameters (weights) across all time steps. Training is performed using *Backpropagation Through Time* (BPTT), which unrolls the network graph for the duration of the sequence.

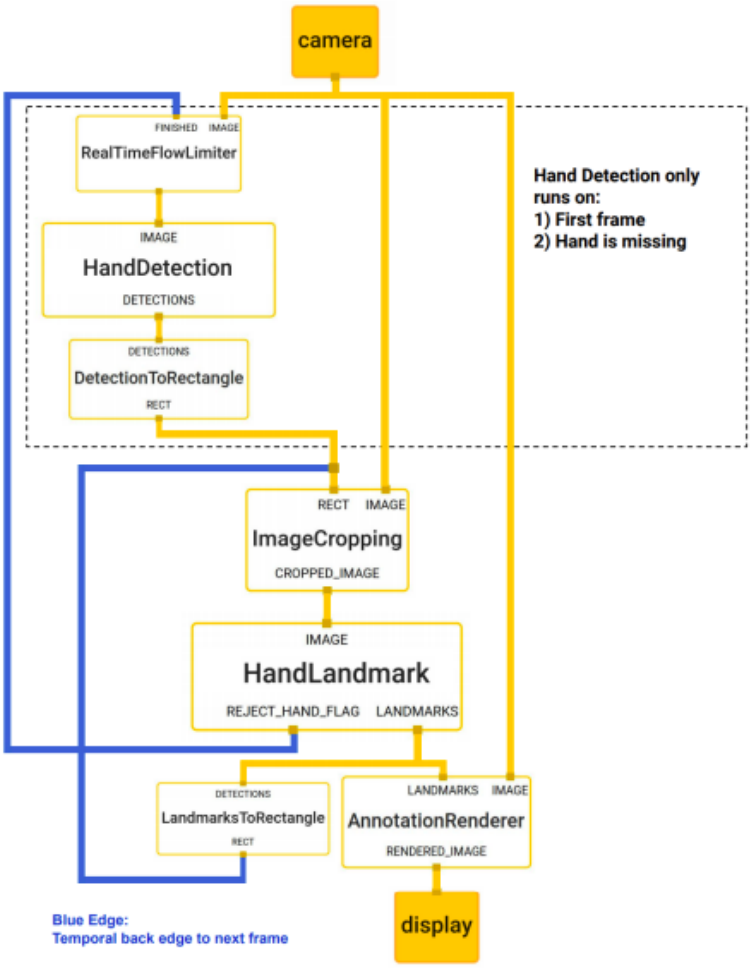


Figure 2.2: MediaPipe Hands pipeline showing the interaction between the detection and landmark models. Source: Zhang et al. [4].

However, RNNs suffer from a critical flaw known as the *vanishing gradient problem*. During BPTT, gradients are calculated by the chain rule, which involves repeated multiplication of the weight matrix. If the eigenvalues of this matrix are smaller than 1, the gradients decay exponentially as they propagate back to earlier time steps. Consequently, the network fails to learn long-term dependencies, effectively "forgetting" the beginning of a sign by the time it reaches the end.

2.3.2 Long Short-Term Memory (LSTM)

To overcome the vanishing gradient problem, Hochreiter and Schmidhuber [5] proposed the Long Short-Term Memory (LSTM) network. The core innovation of the LSTM is the separation of the

memory into two distinct vectors: the *Hidden State* ($\mathbf{h}_t$), which carries short-term information, and the *Cell State* ($\mathbf{C}_t$), which acts as a "highway" for information flow.

The Cell State runs down the entire chain with only minor linear interactions, making it easy for information to flow unchanged. This flow is regulated by three distinct "gates" composed of sigmoid neural network layers:

- **Forget Gate:** Decides what information to discard from the cell state (outputs numbers between 0 and 1).
- **Input Gate:** Decides what new values will be updated in the cell state.
- **Output Gate:** Controls what parts of the cell state are output to the hidden state.

This mechanism, known as the *Constant Error Carousel* (CEC), protects the error signal from decaying, enabling LSTMs to bridge time lags of more than 1000 steps. While highly effective, the complexity of calculating four distinct linear layers for each time step results in high computational costs and parameter counts. The internal structure of an LSTM cell is shown in **Figure 2.3(a)**.

2.3.3 Gated Recurrent Units (GRU)

Gated Recurrent Units (GRUs), proposed by Chung et al. [6], represent a modern simplification of the LSTM architecture. GRUs address the redundancy in LSTMs by removing the separate cell state and merging the input and forget gates into a single **Update Gate** ($\mathbf{z}_t$). They also introduce a **Reset Gate** ($\mathbf{r}_t$).

The mathematical formulation for a GRU at time t is:

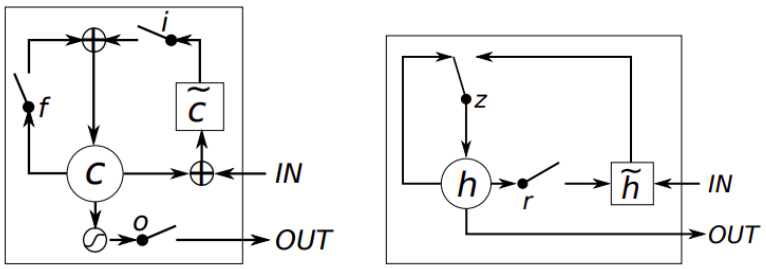
$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \tag{2.3}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \tag{2.4}$$

The candidate hidden state $\tilde{\mathbf{h}}_t$ uses the reset gate to drop irrelevant past information, while the final hidden state is a linear interpolation controlled by the update gate:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{2.5}$$

This interpolation is the key to GRU's efficiency: the network can decide to copy the previous state $\mathbf{h}_{t-1}$ directly (when $\mathbf{z}_t \approx 0$), preserving long-term memory without the complex machinery of the LSTM. Illustrated in **Figure 2.3(b)**, GRUs typically match LSTM performance with fewer parameters, making them less prone to overfitting on datasets with limited samples.



(a) Long Short-Term Memory (b) Gated Recurrent Unit

Figure 2.3: Comparison of Recurrent Units. (a) Long Short-Term Memory (LSTM) with three gates. (b) Gated Recurrent Unit (GRU) with two gates. Source: Chung et al. [6].

2.4 Attention Mechanisms

Attention mechanisms emerged as a response to the "information bottleneck" in fixed-length sequence encodings, first popularized in neural machine translation by Bahdanau et al. [7]. In traditional Recurrent architectures, the entire input sequence must be compressed into a single context vector $\mathbf{v}$. For long sequences, this compression results in loss of detail, as the vector size is fixed regardless of the input length.

Attention solves this by allowing the model to dynamically search for parts of the source sequence that are relevant to the current prediction. Instead of a static vector, Attention computes a *Context Vector* as a weighted combination of all encoder states $\{\mathbf{h}_t\}_{t=1}^{T}$.

Formally, the system learns an alignment model (or scoring function) $a(\mathbf{s}, \mathbf{h}_t)$ that measures how well the inputs around position t and the output at position i match. These scores are normalized using a Softmax function to produce attention weights α_t:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}, \quad e_t = a(\mathbf{s}, \mathbf{h}_t) \tag{2.6}$$

The context vector $\mathbf{c}$ is then derived as:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \mathbf{h}_t \tag{2.7}$$

In the context of SLR, temporal attention acts as a learned filter. Sign language videos often contain "non-discriminative" frames, such as the hands returning to a resting position or preparatory movements. By assigning near-zero weights ($\alpha_t \approx 0$) to these frames, the model effectively ignores the noise and focuses its computational capacity on the "key frames" where the semantic meaning of the sign is contained. This improves robustness to variable execution speeds and intra-class variability.

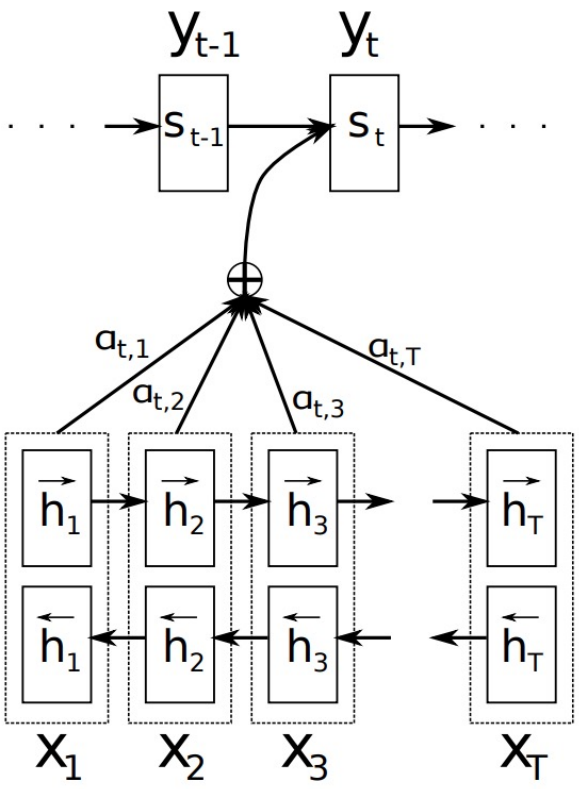


Figure 2.4: Neural Machine Translation architecture with Attention mechanism. The context vector is a weighted sum of encoder annotations. Source: Bahdanau et al. [7].

Chapter 3

Objectives

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3.1 Main objective

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3.2 Specific objectives

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3.3 Validation methods

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Chapter 4

Methodology and Work Plan Overview

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4.1 Materials and Resources

4.2 Methods

4.3 Methodology

4.4 Work Plan

4.5 Constraints and Limitations

Chapter 5

Technical Development

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5.1 System design

Figure 5.1 shows the class diagram.

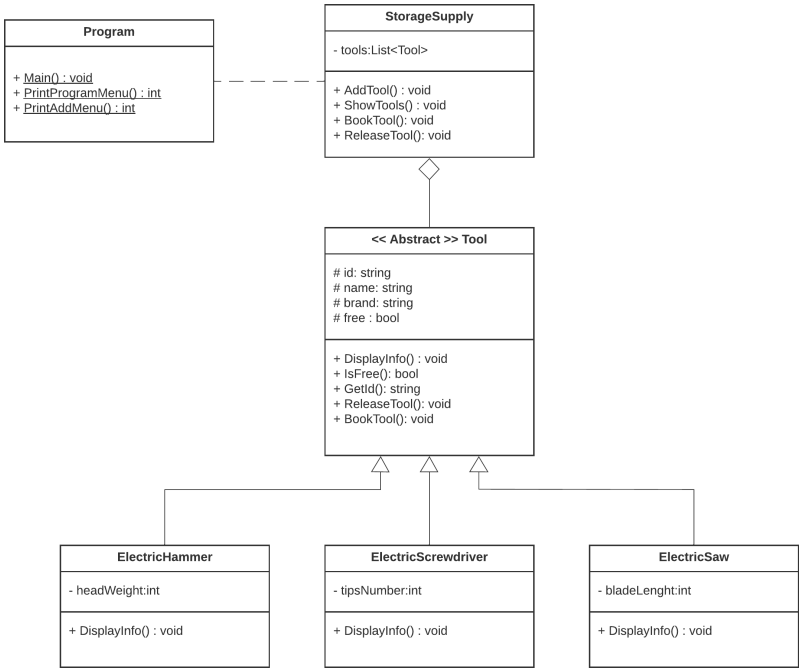


Figure 5.1: Class diagram example.

5.1.1 Description of functional features

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5.1.2 Operational Environment

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5.2 Requirements

5.2.1 Description of requirement attributes

For the textual description of the different identified requirements, a set of attributes has been selected to describe each requirement. Below is a description of the meaning of each attribute used:

- **Code**: A unique abbreviated identifier for the requirement. It is constructed using the requirement type code followed by a hyphen and three digits. Requirements are divided into functional and non-functional requirements. The codes used are **FR** for Functional Requirements and **NFR** for Non-Functional Requirements. For example: `FR-001`.
- **Name**: Extended identification of the requirement.
- **Description**: A basic description of the identified requirement.
- **Source**: Indicates the source through which the requirement was identified. This value usually corresponds to one or more use case codes.
- **Necessity**: Determines the degree of implementation required for the requirement. The possible values are:
  - **Essential**: The requirement must be implemented.
  - **Desirable**: It is preferable to implement the requirement, but it is not mandatory.
  - **Optional**: The requirement may be implemented, but it is neither important nor mandatory.
- **Priority**: Defines the importance of the requirement, allowing the order in which requirements are included in the design and implementation process to be established. The possible values are:
  - **High**: The requirement must be implemented in the early stages of development.

  - **Medium**: The requirement should be implemented once high-priority requirements have been completed.
  - **Low**: The requirement should be implemented in the final stages of development. These requirements do not affect the correct functioning of the system.
- **Stability**: Defines the stability of the requirement throughout the software life cycle. This indicates whether the requirement may be modified during the life cycle. The possible values are:
  - **Stable**: The requirement will not change during the system life cycle.
  - **Unstable**: The requirement may change during the system life cycle.
- **Verifiability**: Defines the degree to which a requirement can be verified, that is, the extent to which it is possible to confirm that the requirement has been implemented in the developed system. The possible values are:
  - **High**: It is possible to verify that the requirement has been implemented. These requirements usually correspond to core system functionalities.
  - **Medium**: It is possible to verify that the requirement has been implemented, but it requires complex testing or inspection of the source code.
  - **Low**: It is difficult, or in some cases impossible, to verify whether the requirement has been implemented.

System requirement			
ID	111	**Source**	111
Name	111		
Description	111		
Necessity	111	**Priority**	111
Stability	111	**Verifiability**	111

Table 5.1: Functional requirement 1

5.3 Datasets

This project spans two recognition tasks with fundamentally different data requirements: hand localization from still images, and word-level sign language recognition from video. As a consequence, five distinct datasets are employed across the full experimental pipeline. This section formalizes the properties of each dataset by first defining a common set of descriptive attributes, then presenting a unified comparative table, and finally providing individual descriptions of each dataset in context.

5.3.1 Dataset Attribute Definitions

To ensure a consistent and unambiguous characterization of every dataset, the following attributes are used in Table 5.2. Each attribute captures a distinct dimension relevant to dataset selection, reproducibility, and the interpretation of experimental results:

- **ID**: A unique alphanumeric identifier assigned to each dataset within this thesis. The prefix `DS-` followed by a two-digit index allows unambiguous cross-referencing in subsequent chapters when reporting experimental configurations and results.
- **Name**: The official or widely-used name of the dataset as it appears in the literature or on its hosting platform.
- **Year**: The year of original publication or first public release. This contextualizes the dataset within the chronological development of the field and helps evaluate whether it reflects the current state of the art in data collection methodology.
- **Task**: The specific sub-task for which the dataset was designed. Three task categories appear in this project:
  - **Hand Detection**: the dataset provides annotated bounding boxes or segmentation masks marking the location of hands in images.
  - **Letter Recognition**: the dataset provides labeled static images of individual American Sign Language (ASL) letter signs.
  - **Word Recognition**: the dataset provides labeled video clips of complete ASL word signs, requiring temporal modeling.
- **Samples**: The total number of annotated data instances. For image datasets, this equals the number of images; for video datasets, it equals the number of individual video clips. Where a subset of the full dataset was used in this project, this is noted in the individual descriptions.
- **Classes**: The number of distinct output categories. For hand detection datasets this is always 1 (the single *hand* class); for sign language datasets it corresponds to vocabulary size — the number of distinct letters or word glosses.
- **Format**: The storage format and annotation type of the raw data. This attribute specifies whether instances are static images or video clips, the image resolution where relevant, and the form of ground-truth annotation (e.g., bounding box coordinates in XML, pixel-level segmentation masks, image class labels, or video-level class labels).
- **Input Representation**: Whether the models trained on this dataset consume *pixel-based* data (raw RGB or grayscale image/video frames) or *landmark-based* data (structured 3D keypoint coordinates extracted by a pose estimator such as MediaPipe). Pixel-based inputs preserve full visual detail but are high-dimensional and sensitive to background, lighting, and signer appearance. Landmark-based inputs are compact, viewpoint-invariant, and more robust to visual variability, at the cost of depending on a reliable keypoint extractor.

5.3.2 Comparative Overview

Table 5.2 provides a high-level comparison of the five datasets used in this project. Datasets DS-01 and DS-02 support the hand localization pipeline described in later experimental sections. Datasets DS-03 and DS-04 were used in the initial letter-level classification phase of the project, which was subsequently superseded by the word recognition approach. Dataset DS-05 is the primary benchmark for all word-level recognition experiments and constitutes the main evaluation resource throughout the thesis.

Table 5.2: Comparative overview of all datasets used in this project.

ID	Name	Year	Task	Samples	Classes	Format
DS-01	EgoHands	2015	Hand Detection	4,800	1	Images + Seg. Masks
DS-02	Kaggle Hand Dataset	2023	Hand Detection	∼4,200	1	Images + BBox (XML)
DS-03	Sign Language MNIST	2017	Letter Recog.	34,627	24	28×28 Grayscale CSV
DS-04	ASL Alphabet	2018	Letter Recog.	87,000	29	200×200 RGB JPG
DS-05	WLASL	2020	Word Recog.	21,083	2,000	Video Clips (RGB)

5.3.3 Hand Detection Datasets

Datasets DS-01 and DS-02 are used jointly to train the hand localization module based on YOLOv8. A central design motivation for combining two datasets is improved generalization: a detector trained exclusively on egocentric footage tends to fail when the camera is positioned at eye level or beyond arm's reach, while a detector trained only on third-person images may struggle with first-person perspectives. By merging the two sources, the training corpus covers a significantly wider range of viewpoints, backgrounds, and hand appearances. After merging, duplicate removal, and a stratified train/validation/test split, the combined training set comprises approximately 5,223 annotated images.

DS-01 — EgoHands [8]

EgoHands is an egocentric hand detection dataset published by Indiana University in 2015 as part of research on activity recognition in close-proximity social interactions. It consists of approximately 4,800 labeled frames extracted from 48 video recordings captured from a GoPro camera mounted on the chest of an observer watching another person. Ground-truth annotations are stored as pixel-level segmentation polygons in MATLAB `.mat` files; during preprocessing, these polygons were converted to axis-aligned bounding boxes in YOLO format. The dataset covers diverse interaction scenarios — including card games, chess, and jigsaw puzzles — with two hands simultaneously visible in most frames, providing challenging overlap and occlusion cases. Because all images are taken from a first-person perspective, the visual distribution of EgoHands is inherently biased toward the observer's own hand. This limitation motivates the inclusion of DS-02 to extend viewpoint diversity. The input representation is pixel-based: models consume raw RGB crops.

DS-02 — Kaggle Hand Dataset [9]

The Kaggle Hand Dataset is a collection of annotated hand images published on the Kaggle platform, providing multi-viewpoint coverage that is complementary to the egocentric nature of DS-01. Images depict hands in third-person perspectives with diverse backgrounds, lighting conditions, skin tones, and partial occlusions. Ground-truth bounding boxes are provided in Pascal VOC XML format; during preprocessing, coordinates were normalized to the YOLO format used by the detection pipeline. The dataset contains approximately 4,200 labeled images. The joint training corpus formed by DS-01 and DS-02 yielded a YOLOv8n model achieving approximately 98% mAP50 and 87%

Mean IoU on the held-out test split, demonstrating the effectiveness of the multi-source approach. The input representation is pixel-based.

5.3.4 Letter Recognition Datasets

Datasets DS-03 and DS-04 were used during the initial phase of the project, when the recognition target was the static letter-level ASL alphabet. Both datasets cover 24 of the 26 ASL letters; the letters *J* and *Z* are excluded in both cases because they require hand motion and cannot be represented by a single static image. The two sources were merged and unified into a single training corpus for letter classification experiments. Although this phase was later superseded by the more ambitious word-level recognition approach, these datasets remain relevant because they established the viability of data-driven letter classification and informed the design of the preprocessing pipeline. The input representation for both datasets is pixel-based.

DS-03 — Sign Language MNIST [10]

Sign Language MNIST mirrors the format of the canonical MNIST handwritten digit benchmark. It contains 27,455 training images and 7,172 test images — 34,627 samples in total — each representing one of 24 static ASL letters as a 28×28 grayscale image. Data is distributed in CSV format, with one row per image where each column corresponds to a pixel intensity value in the range $[0, 255]$, plus a label column. The low resolution and grayscale format make this dataset extremely lightweight and straightforward to preprocess, but inherently limit the discriminative information available to the model, as fine-grained finger articulation details are largely lost. The dataset was originally constructed from ASL alphabet photographs rescaled and converted to the MNIST-style format, and is hosted on Kaggle.

DS-04 — ASL Alphabet [11]

The ASL Alphabet dataset provides a substantially richer visual input than DS-03. It contains approximately 87,000 images organized into 29 classes: the 24 static ASL letters, the letter *space*, and two additional non-letter classes (*del*, *nothing*). Each class is represented by exactly 3,000 training images captured in a controlled environment at a resolution of 200×200 pixels in full RGB color. The higher resolution and color information enable more discriminative feature learning, particularly for hand texture and illumination cues. For integration with DS-03 into a unified corpus, images were converted to grayscale and resized to a common format, yielding a combined training set of approximately 79,000 images and 7,000 test images covering the 24 shared letter classes.

5.3.5 Word Recognition Dataset

DS-05 — WLASL [12]

The Word-Level American Sign Language (WLASL) dataset, published by Li et al. in 2020, is the primary data resource for all word-level recognition experiments in this thesis and one of the largest publicly available ASL word recognition benchmarks to date. WLASL contains 21,083 video instances covering 2,000 ASL glosses (words). Videos were collected from multiple online platforms — including YouTube channels maintained by ASL educators, online courses, and signing dictionaries — and manually verified by qualified ASL experts to ensure correctness of sign production. Each clip shows a single signer performing one complete sign, typically against a plain or low-clutter background, with durations ranging from approximately one to four seconds.

The dataset is partitioned into four progressively larger evaluation subsets: WLASL100 (100 glosses), WLASL300 (300 glosses), WLASL1000 (1,000 glosses), and WLASL2000 (the full 2,000 glosses). This structure enables systematic benchmarking of how model performance scales with vocabulary size. A practical challenge is that WLASL relies on external video URLs; a fraction of clips have become unavailable over time due to deletion or access restriction, so only successfully downloaded videos can be used.

In this project, raw RGB frames are not used as model input. Instead, MediaPipe Hands [4] is applied to each frame to extract 21 three-dimensional hand keypoints, yielding a compact landmark vector of 63 scalar features per frame (21 joints $\times$ 3 coordinates). This representation reduces input dimensionality by a factor of approximately 2,389 compared to raw 224×224 RGB frames, removes background and appearance noise, and provides geometric invariance to signer appearance. The effective detection rate of MediaPipe on WLASL clips is approximately 69%; frames where no hand is detected are linearly interpolated from adjacent detected frames to maintain sequence continuity. Each video is resampled to a fixed sequence length of 64 frames before being fed to the model.

For the experiments described in this thesis, the following WLASL subsets were extracted and used:

- **WLASL30**: The 30 most frequent glosses, used for initial architecture exploration and model selection. Split: 242 training / 56 validation / 34 test videos.
- **WLASL100**: The 100 most frequent glosses, used for K-Fold cross-validation and robust evaluation. Split: 741 training / 162 validation / 99 test videos.
- **WLASL300, WLASL1000, WLASL2000**: Larger subsets targeted for progressive scaling experiments in later stages of the project, where the goal is to assess model accuracy and generalization at increasing vocabulary sizes.

Chapter 6

Experiments

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Chapter 7

Ethical Implications and Social Impact

7.1 AI-Based Systems Ethical Impact

7.2 Ethical Dimensions Descriptions

7.3 Project Analysis

7.4 Impact scale of Ethical Aspects

Chapter 8

My Journey at UFV

Chapter 9

Conclusiones

9.1 General Conclusions

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9.3 Next Steps

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